

BIOL 4605 / 7220 Name _____ circle one 4605: (no project) or 7220
OCSC 4605 / 7220 18 September 2026
Quiz 1

- You may refer to paper notes and textbooks to complete this Quiz.
- You may not use the internet or access mobile phones or computers.
- You have 15 minutes to complete this quiz.
- You may use a calculator. If you do not have one, you may ask the instructor to complete a calculation

1a. Complete the following tables, where $Y = E[Y] + \varepsilon$ [3 marks]:

Model 1: $E[Y] = 10$

X	Y	$E[Y]$	ε
10	8		
20	12		

Model 2: $E[Y] = 0.5X + 4$

X	Y	$E[Y]$	ε
10	8		
20	12		

2b. What is the residual sum of squares for Model 1? [1 mark]

2c. What is the residual sum of squares for Model 2? [1 mark]

3. Consider the data equation:

$$Y = \beta_1 X + \beta_0 + \epsilon$$

where Y is particulate matter in the air (units: grams/m³) and X is elevation (km). Complete the sentences below:

- a. The units of β_1 are _____ [1 mark]
- b. The units of β_0 are _____ [1 mark]
- c. The name (in words) for the symbol ϵ _____ [1 mark]
- d. The units of ϵ are _____ [1 mark]

4a. Sketch a graph with X on the horizontal axis and Y on the vertical axis. Add:

- data (dots) showing a linearly increasing relationship between Y and X. [1 mark]
- a dashed line that shows no relationship between X and Y. [1 mark]
- an approximate best fit line to your data. [1 mark]

4b. What is the slope of the dashed line? _____ [1 mark]

4c. What is true of the best fit line that you drew (in contrast to your answer for 4b)?

_____ [1 mark]